**Microwave Engineering- ECA1308**

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**Date:15-102026**

**EXPERIMENT 2(B):**

Determination of Optimum Operating Frequency from S-Parameter Sweep

**Aim**

To determine the optimum operating frequency from the 8–12 GHz S-parameter sweep by finding the frequency at which $|S_{11}|$ is minimum and $|S_{21}|$ is maximum, and to compare its performance with that at 10 GHz.

**Software Required:**

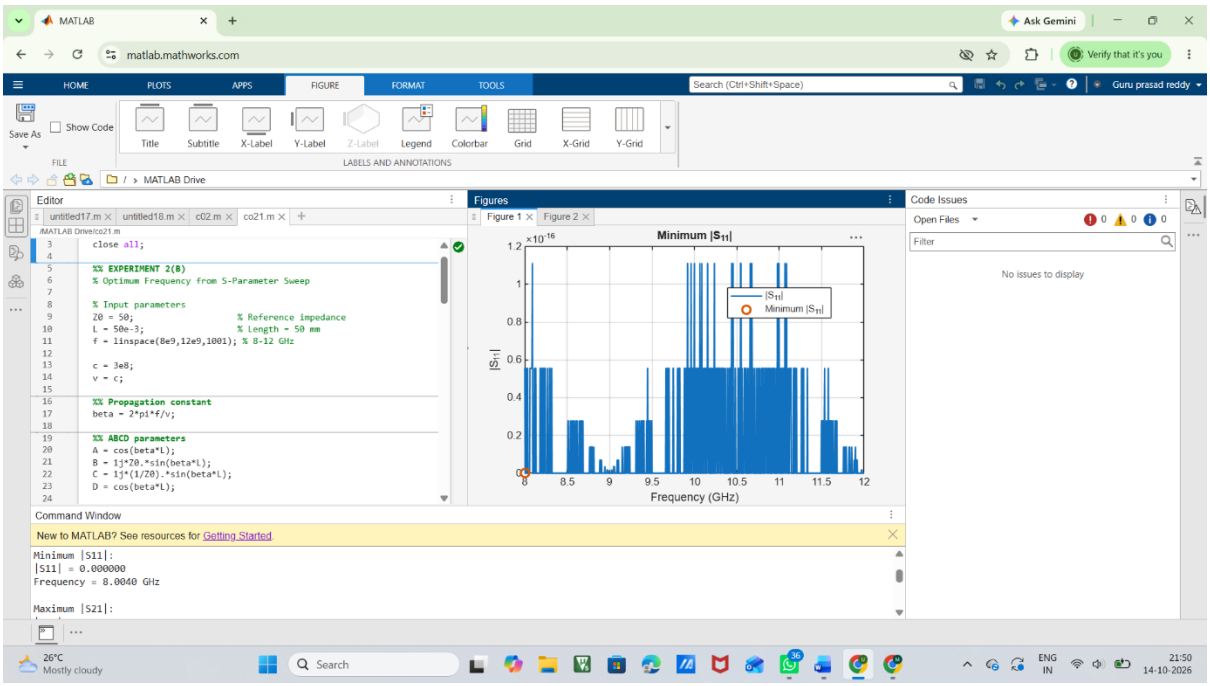
- MATLAB

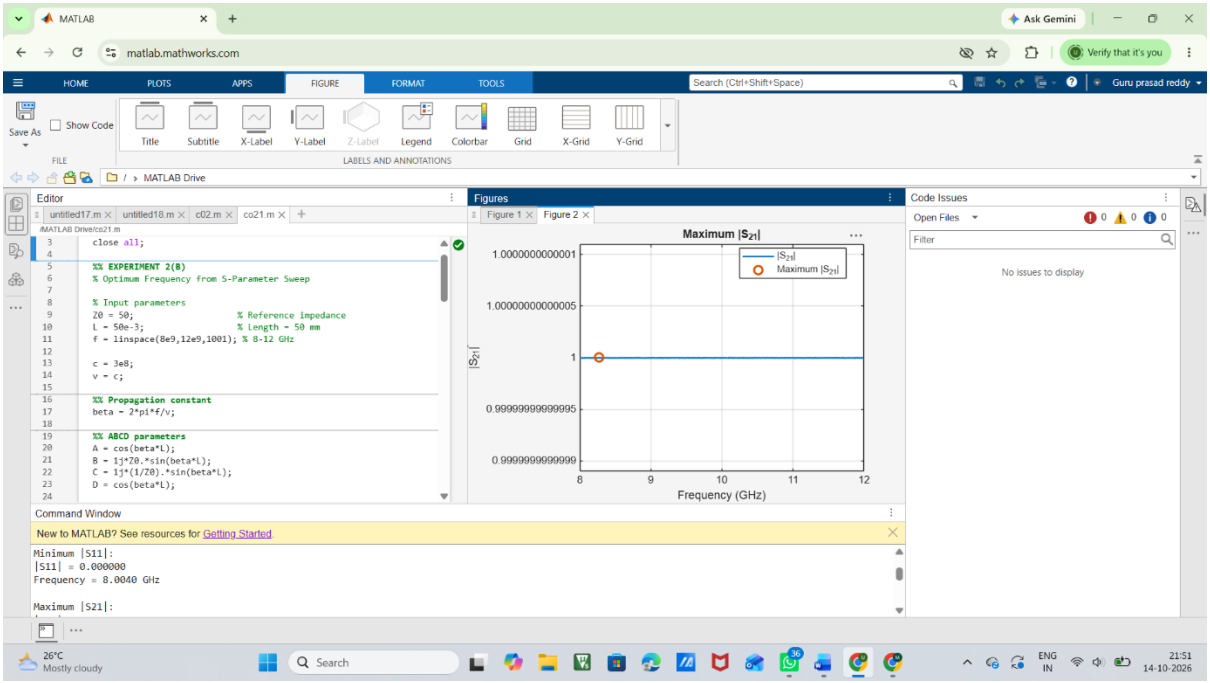
**Procedure — MATLAB Process**

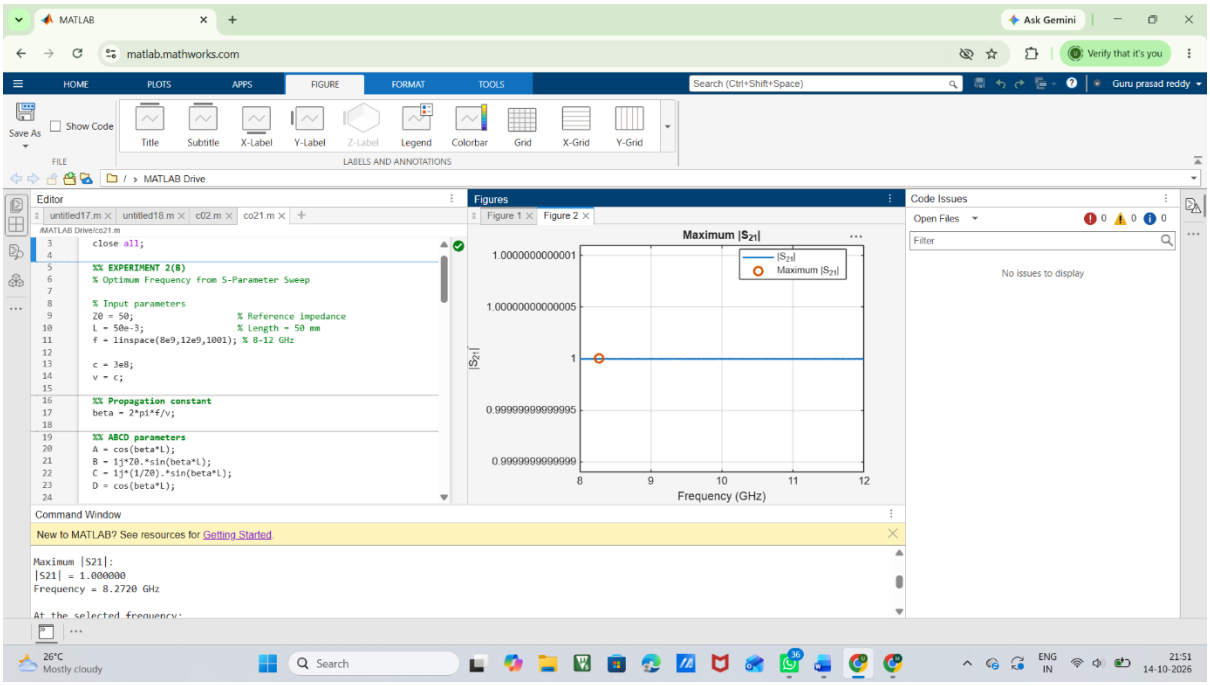
1. Open MATLAB and create a new script for the optimum-frequency analysis.
2. Enter the transmission-line parameters: $Z_0=50\ \Omega$, $L=50$ mm, and frequency range from 8 GHz to 12 GHz.
3. Generate the frequency sweep using linspace(8e9,12e9,1001).
4. Calculate the propagation constant $\beta$ for every frequency in the sweep.
5. Calculate the ABCD parameters $A,B,C,D$ of the ideal lossless 50 Ω transmission line.
6. Convert the ABCD parameters into $S_{11}$ and $S_{21}$ over the complete frequency range.
7. Calculate the magnitudes $|S_{11}|$ and $|S_{21}|$ using the MATLAB abs() function.
8. Use the min() function to find the minimum $|S_{11}|$ and the max() function to find the maximum $|S_{21}|$, along with their corresponding frequencies.

9. Calculate the return loss and insertion loss at the selected frequency and plot \(|S_{11}|\) and \(|S_{21}|\) against frequency to identify the optimum operating condition.

**OUTPUT:**






**Result —**

1. The minimum \(|S_{11}|\) and maximum \(|S_{21}|\) were successfully determined from the 8–12 GHz sweep.
2. For the ideal 50 Ω matched transmission line, \(|S_{11}|\approx0\) throughout the frequency range.
3. Similarly, \(|S_{21}|\approx1\) throughout the frequency range, giving nearly 100% transmission efficiency.
4. Therefore, there is no unique optimum frequency; all frequencies from 8 to 12 GHz provide the same ideal performance.
5. The selected frequency gives excellent matching and transmission because both ports are perfectly matched to the 50 Ω characteristic impedance.